

Module 2 Review

Gauss Jordan elimination

The three valid moves we can make are:

1. **Multiplication/Division by a scalar** – Multiply a row by a non-zero scalar
2. **Row combination** - Replace a row with the sum/difference of that row and a (nonzero) multiple of another row.
3. **Swapping** - Swap two rows.

The number of equations in the linear system is the number of rows r in the augmented matrix, and the number of columns k in the matrix left of the vertical line is the number of unknowns.

Reduced Row Echelon Form (RREF)

A matrix is said to be in RREF if it has the following properties:

1. The first non-zero element in each row is called the *Pivot* of that row. In RREF. The pivot of each row is 1.
2. The pivot of any given row is always to the right of the pivot of the row above it.
3. The pivot is the only non-zero entry in its column.

Solution set of a system of linear equations:

- If Gauss-Jordan elimination results in a row of all zeros, except for the number on the right-hand side of the augmented matrix, then there are no solutions.

- If Gauss Jordan Elimination leads to a row of all 0's on either side of the Augmented Matrix and the number of remaining non zero rows is less than the number of unknowns, then it's the case of infinite solutions. Such systems are called *dependent systems*.
- When Gauss-Jordan elimination results in ones along the diagonal on the left-hand side, one per variable, and zeros elsewhere on that side (Identity Matrix), we have the case of unique solutions.

A square matrix that leads to a unique solution is called a Non-Singular matrix while the ones that do not lead to unique solutions are called Singular Matrices.

Length of a vector:

$$|\vec{v}| = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}$$

A vector with magnitude 1 is called a Unit Vector

Dot Product of two vectors: $\rightarrow$ always a scalar

$$\vec{a} \cdot \vec{b} = a_1 \cdot b_1 + a_2 \cdot b_2 \dots + a_n \cdot b_n$$

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$

Vectors which have an angle of 90° between them are called **Orthogonal** vectors.

Matrix Multiplication:

Let $\mathbf{A} = (a_{ij})$ and $\mathbf{B} = (b_{ij})$, if the number of columns of $\mathbf{A}$ is the same as number of rows of $\mathbf{B}$, then the product of $\mathbf{A}$ and $\mathbf{B}$ is given by $\mathbf{AB} = (c_{ij})$ where c_{ij} is dot product of i^{th} row of $\mathbf{A}$ with j^{th} column of $\mathbf{B}$.

Vector Space

Consists of a set of vectors and two operations: addition of vectors and multiplication by a scalar.

To check validity of a vector space, with a defined set of vectors but traditional vector addition and scalar multiplication, check for existence of zero vector and closeness under addition and multiplication.

For custom vector addition and scalar multiplication, the following axioms need to be checked (will be given):

- $\vec{x} + \vec{y} = \vec{y} + \vec{x}$
- $(\vec{x} + \vec{y}) + \vec{z} = \vec{x} + (\vec{y} + \vec{z})$
- There exists some vector $\vec{0} \in V$, such that $\vec{x} + \vec{0} = \vec{x}$
- For each vector $\vec{x} \in V$, there exists a unique vector $\overline{\vec{x}} \in V$, such that $\overline{\vec{x}} + \vec{x} = \vec{0}$
- $a(\vec{x} + \vec{y}) = a\vec{x} + a\vec{y}$
- $(a+b)\vec{x} = a\vec{x} + b\vec{x}$
- $a(b\vec{x}) = (ab)\vec{x}$
- $1\vec{x} = \vec{x}$.

Vector Subspace

To check validity of a non-empty subspace W which is a subset of space V , check for presence of zero vector and closeness under vector addition and scalar multiplication.

Linear Combination

A vector $\vec{y}$ in vector space V (over $\mathbb{R}$) is a linear combination of the vectors $\vec{x}_1, \vec{x}_2, \dots, \vec{x}_n$ if there exist scalars $a_1, a_2, \dots, a_n$ in $\mathbb{R}$ such that $\vec{y} = a_1\vec{x}_1 + a_2\vec{x}_2 + \dots + a_n\vec{x}_n$

Vector Space Generating Set

The **span** of a set of vectors is defined as the collection of all possible linear combinations of those vectors.

When a set of vectors spans an entire Vector Space, then it is called a Vector Space Generating Set.

Linear Independence

A set of vectors $\{\vec{x}_1, \vec{x}_2, \dots, \vec{x}_n\}$ from a real vector space V is linearly independent if the only solution to the equation $a_1\vec{x}_1 + a_2\vec{x}_2 + \dots + a_n\vec{x}_n = \vec{0}$ is

$a_1 = a_2 = a_3 = \dots = a_n = 0$. Otherwise, the set is called a linearly dependent set.

Basis

A basis for a vector space is a set of vectors that is linearly independent and generated the vector space.

Let V be a subspace of $\mathbb{R}^n$. A **basis** of V is a set of vectors $\{\vec{x}_1, \vec{x}_2, \dots, \vec{x}_m\}$ in V such that:

1. $V = \text{Span}\{\vec{x}_1, \vec{x}_2, \dots, \vec{x}_m\}$, and
2. the set $\{\vec{x}_1, \vec{x}_2, \dots, \vec{x}_m\}$ is linearly independent.

The **dimension** of a space is the number of basis vectors required to span it.

Generally, the standard co-ordinate vectors:

$$\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \vec{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \dots, \vec{e}_n = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix} \text{ form a basis for } \mathbb{R}^n. \text{ This is known as the } \textit{standard}$$

basis.

Change of Basis

If we have two bases:

1. $\{\vec{x}_1, \vec{x}_2, \dots, \vec{x}_m\}$ and

2. $\{\vec{y}_1, \vec{y}_2, \dots, \vec{y}_m\}$

and we are given the co-ordinates of a vector $\vec{v}$ in basis 1 and we want to find the co-ordinates of $\vec{v}$, say $\vec{v}'$, in basis 2, we use:

$$v_1 \vec{x}_1 + v_2 \vec{x}_2 + \dots + v_m \vec{x}_m = v'_1 \vec{y}_1 + v'_2 \vec{y}_2 + \dots + v'_m \vec{y}_m$$

Rank

The *Rank* of a Matrix is a fundamental concept in linear algebra that measures the **number of linearly independent rows or columns in a matrix.**

The Rank of the matrix becomes the number of non-zero rows remaining after applying Gauss Jordan and reaching REF.

Row rank is always equal to the column rank.

If the rank of a matrix is equal to the number of unknowns (columns), then the matrix is nonsingular, and if a solution exists, it will be unique.

Transpose of a matrix

The transpose of a matrix A is represented by A^T and is obtained by swapping the rows and columns of the matrix.

Some important properties of the transpose of a matrix:

- A square matrix “ A ” of order “ $n \times n$ ” is said to be ***orthogonal matrix***, if $AA^T = A^T A = I$, where “ I ” is an identity matrix of order “ $n \times n$.”
- A square matrix “ A ” of order “ $n \times n$ ” is said to be a ***symmetric matrix*** if its transpose is the same as the original matrix, i.e., $A^T = A$.
- A square matrix “ A ” of order “ $n \times n$ ” is said to be a ***skew-symmetric matrix*** if its transpose is equal to the negative of the original matrix, i.e., $A^T = -A$.
- ***Double Transpose of a Matrix:*** Transpose of the transpose matrix is the original matrix itself.

$$(A^T)^T = A$$

- ***Transpose of product of Matrices:*** This property says:

$$(AB)^T = B^T A^T$$

- ***Transpose of addition of Matrices:*** This property says:

$$(A + B)^T = A^T + B^T$$

Determinant of a Matrix

The determinant of a matrix is a scalar value that can be calculated for a square matrix and gives us information about the singularity of matrix and is *zero iff A is singular* (and not invertible).

Minor of element of Matrix:

Minors are the determinants for every element of a matrix obtained by eliminating the rows and columns of that element. So, if we have a matrix:

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Then the minor of a_{12} will be given by M_{12} :

$$\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}$$

Co-factor of element of a matrix:

For an element a_{ij} , the cofactor of this element is C_{ij} and the minor is M_{ij} then, the cofactor can be written as:

$$C_{ij} = (-1)^{i+j} M_{ij}$$

The determinant of a Matrix is calculated along any one row or column of a Matrix by adding the multiplying the elements of the row/column to their co-factors and adding them. It is always easier to choose the row/column with the maximum number of zeroes to reduce the amount of computation.

Inverse of a Matrix

The **inverse of a Matrix “A”**, denoted as A^{-1} , is **another matrix that, when multiplied by A**, results in the **identity matrix I**. In mathematical terms:

$$A \times A^{-1} = A^{-1} \times A = I$$

A matrix must be **square** (same number of rows and columns) and must be **non-singular** (its determinant is not zero) to have an inverse.

$$A^{-1} = \frac{\text{Adj}(A)}{\text{Det}(A)}$$

$\text{Adj}(A)$ is the **transpose of the matrix $[A_{ij}]_{n \times n}$** , where A_{ij} is the **cofactor of the element a_{ij}** .

You can also use **Elementary transformation** method to find the inverse of a matrix.

Step 1: Write the given matrix as $A = IA$, where I is the *identity matrix* of the order/dimensions same as A.

Step 2: Use the sequence of *valid row operations* till we get an identity matrix on the LHS (keep applying the same elementary operations on the RHS). We will end up getting the following equation: $I = BA$. We know that $AA^{-1} = I$. So, **matrix B on RHS is the inverse of matrix A**.

For a square matrix: Rank <number of columns -> singular -> no unique solutions ->not invertible-> determinant is 0.

Vector projection of one vector onto another

If we have two vectors $\vec{v} = (v_1, v_2)$ and $\vec{u} = (u_1, u_2)$ and we want to project $\vec{u}$ on $\vec{v}$. We write it as: **$Proj_{\vec{v}} \vec{u}$** and is given by:

$$Proj_{\vec{v}} \vec{u} = \frac{\vec{u} \cdot \vec{v}}{|\vec{v}|^2} \vec{v} = \frac{\vec{u} \cdot \vec{v}}{\vec{v} \cdot \vec{v}} \vec{v}$$

Projection in higher dimensions:

$$Proj_M \vec{v} = A(A^T A)^{-1} A^T \vec{v}$$

Here A is the matrix containing the basis vectors of the space we are projecting onto.

Line of best fit:

$$y = mx$$

$$Proj_{\vec{x}} \vec{y} = \frac{\vec{y} \cdot \vec{x}}{\vec{x} \cdot \vec{x}} \vec{x} = m \vec{x}$$

$$y = mx + b$$

$$y = m_1 x_1 + m_2 x_2 + \dots + m_n x_n + b$$

$$y = m_1 x^1 + m_2 x^2 + \dots + m_n x^n + b$$

$$\begin{bmatrix} \vec{x}_1 & \vec{x}_2 & \dots & \vec{x}_n & 1 \\ \vdots & \vdots & & \vdots & \vdots \end{bmatrix}$$

$$\vec{c} = \begin{bmatrix} m_1 \\ m_2 \\ m_3 \\ \vdots \\ b \end{bmatrix}$$

$$\vec{c} = (A^T A)^{-1} A^T \vec{y}$$

Here $\vec{c}$ is a vector containing the coefficient of x's and the intercept and A is a matrix containing all the predictor columns plus the column of 1's.

Let's practice

1. Use Gauss–Jordan to determine if the matrix is invertible:

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 3 \\ -1 & 1 & 0 \end{bmatrix}$$

(If invertible, compute A^{-1} .)

2. Let A be the matrix

$$A = \begin{bmatrix} 1 & 2 & a \\ 2 & 4 & b \\ 3 & 6 & c \end{bmatrix}$$

Find all values of a, b, c for which the RREF has:

- (a) rank 1
- (b) rank 2
- (c) rank 3

3. For the matrix

$$A = \begin{bmatrix} 2 & -1 & 3 \\ 4 & -2 & 6 \\ 1 & 0 & 1 \end{bmatrix}$$

- (a) Determine the **rank** of A .
- (b) Determine whether A is **invertible**.
- (c) Are the columns **linearly independent**?
- (d) Compute $\det(A)$

4. Is the set

$$S = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 \leq 1\}$$

a valid vector space?

5. A movie-recommendation system represents two users' preferences as feature vectors:

$$u = (4, 1, 5, 0), v = (2, 3, 1, 4)$$

Each entry represents a rating for one genre (action, comedy, drama, horror).

Compute the **dot product** and explain whether the users are **positively aligned** or not.

6. A search engine embeds a query as $q = (0.5, 0.1, 0.7, 0.2)$ and a document as $d = (0.4, 0.2, 0.6, 0.1)$. Compute the projection of d onto q .

This measures how much the document lies in the “direction” of the query.

7. You are given ad spending $x = (1, 2, 3)$ and weekly sales $y = (2, 5, 4)$.

(a) Compute the projection of y onto x .

(b) Use it to compute the best-fit slope m for the model $y = mx$.

(c) Conceptually: What does the sign and magnitude of m say about how well ads predict sales?

Ans:

1. $A = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 3 \\ -1 & 1 & 0 \end{bmatrix}$

QUICK CHECK

The determinant should be non-zero for the matrix to be invertible

$$\det A = 1(1 \times 0 - 3 \times 1) + 1(2 \times 1 - (-1) \times 1) = -3 + 3 = 0$$

$\therefore$ The matrix is NOT invertible

NOW

Using Gauss Jordan :

$$\begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 3 \\ -1 & 1 & 0 \end{bmatrix} \xrightarrow[\substack{R_2 \leftarrow R_2 - 2R_1 \\ R_3 \leftarrow R_3 + R_1}]{} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$\therefore$ Row of all zeros

$\therefore$ NOT invertible

2. Given,

$$A = \begin{bmatrix} 1 & 2 & a \\ 2 & 4 & b \\ 3 & 6 & c \end{bmatrix}$$

a) Rank 1 $\rightarrow$ only 1 linearly independent row/column

$$\text{Column 2} = 2 \times \text{column 1}$$

$$\therefore \text{Column 3} = k \times \text{column 1} \quad (\text{where } k \text{ is any real number})$$

$$\therefore a = k \times 1$$

$$b = k \times 2$$

$$c = k \times 3$$

$$\therefore (a, b, c) = (k, 2k, 3k)$$

b) Rank 2 $\rightarrow$ 2 linearly independent rows/columns

$\therefore$ all values of (a, b, c) such that

$$a \neq k \times 1$$

$$b \neq k \times 2$$

$$c \neq k \times 3$$

$$\therefore (a, b, c) \neq (k, 2k, 3k) \text{ where } k \text{ is any real number}$$

c) Rank 3 $\rightarrow$ 3 linearly independent rows/columns

Since $\text{Column 2} = 2 \times \text{column 1}$, Rank 3 is

NOT possible for any value of (a, b, c)

3. Given

$$A = \begin{bmatrix} 2 & -1 & 3 \\ 4 & -2 & 6 \\ 1 & 0 & 1 \end{bmatrix}$$

Since $R_2 = 2R_1$, Determinant of $A = 0$

which makes A NOT invertible

Since Row Rank = Column Rank, the columns are linearly dependent

Rank < 3

Checking Rank : Converting to REF

$$R_2 = 2R_1 \rightarrow$$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\therefore \text{Rank} = 2$$

4. Given,

$$S = \{ (x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 \leq 1 \}$$

To check :

1. Presence of zero vector
2. Closeness under vector addition
3. Closeness under scalar multiplication

1. zero vector: $(0, 0, 0) \in \mathbb{R}^3$

$$\begin{aligned} x=0 \\ y=0 \\ z=0 \end{aligned} \quad x^2 + y^2 + z^2 = 0 < 1$$

$\therefore$ zero vector is a part of the vector space ✓

2. Closure under vector addition

Let $(x_1, y_1, z_1) \in S$ where $x_1^2 + y_1^2 + z_1^2 = 1$

Let $(x_2, y_2, z_2) \in S$ where $x_2^2 + y_2^2 + z_2^2 = 1$

Adding $(x_1, y_1, z_1) + (x_2, y_2, z_2) =$

$$(x_1 + x_2, y_1 + y_2, z_1 + z_2)$$

$$\begin{aligned} \text{here } (x_1 + x_2)^2 + (y_1 + y_2)^2 + (z_1 + z_2)^2 \\ = x_1^2 + x_2^2 + y_1^2 + y_2^2 + z_1^2 + z_2^2 + 2x_1x_2 + 2y_1y_2 + 2z_1z_2 \end{aligned}$$

$$= 1 + 1 + 2(x_1x_2 + y_1y_2 + z_1z_2)$$

$$> 1 \quad \therefore \notin S \quad \times$$

Eg. $(1, 0, 0) + (0, 0, 1)$

$$= (1, 0, 1)$$

here $1^2 + 0 + 1^2 = 1 + 0 + 1 = 2 > 1 \notin S$

NOT valid vector space

5. Given,

$$\vec{u} = (4, 1, 5, 0)$$

$$\vec{v} = (2, 3, 1, 4)$$

$$\begin{aligned}\vec{u} \cdot \vec{v} &= 4 \times 2 + 1 \times 3 + 5 \times 1 + 0 \times 4 \\ &= 8 + 3 + 5 = 16\end{aligned}$$

Since $\vec{u} \cdot \vec{v} > 0$, The preference vectors are positively aligned $\Rightarrow$ users tend to like similar genres of movies

$$6. \text{Proj}_{\vec{q}} \vec{d} = \frac{\vec{d} \cdot \vec{q}}{\vec{q} \cdot \vec{q}} \vec{q}$$

$$\begin{aligned}\vec{d} \cdot \vec{q} &= 0.5 \times 0.4 + 0.1 \times 0.2 + 0.7 \times 0.6 + 0.2 \times 0.1 \\ &= 0.66\end{aligned}$$

$$\begin{aligned}\vec{q} \cdot \vec{q} &= 0.5^2 + (0.1)^2 + (0.7)^2 + (0.2)^2 \\ &= 0.79\end{aligned}$$

$$\therefore \text{Proj}_{\vec{q}} \vec{d} = \frac{0.66}{0.79} (0.5, 0.1, 0.7, 0.2)$$

$$\approx (0.4177, 0.0835, 0.5848, 0.1671)$$

7.

a) projection of y onto $x = \frac{\vec{y} \cdot \vec{x}}{\vec{x} \cdot \vec{x}} \cdot \vec{x}$

$$= \frac{2 \times 1 + 5 \times 2 + 4 \times 3}{1^2 + 2^2 + 3^2} (1, 2, 3)$$

$\frac{24}{14} = \frac{12}{7}$

$$= \left(\frac{12}{7}, \frac{24}{7}, \frac{36}{7} \right)$$

b) slope, m , for line of best fit $y = mx$

$$m = \frac{\vec{y} \cdot \vec{x}}{\vec{x} \cdot \vec{x}} = \frac{12}{7} \approx 1.71$$

c) Since $m > 0$, ad spending and sales are positively co-related, each unit of ad spending predicts ≈ 1.71 units of sales